Joshua Choong

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Technical Skills

Languages: Python, C, TypeScript, JavaScript, HTML, CSS, Java, LaTeX
Libraries and Frameworks: Scikit-Learn, Flask, Pandas, NumPy, Pennylane, Axios, Vite, Node.js
Tools and Methodologies: Git, Docker, Cursor, Claude Code, Node.js, Jenkins, Agile Standups, Scrum, Jira, Linux

Experience

Systems Developer Co-op @ RLB LLP - Waterloo ON *Sep. 2026 - Dec. 2026*
- Engineered Python automation tools integrating the CCH iFirm tax API to extract and process document data
- Automated annual document rollovers through backend integrations, streamlining error-prone manual workflows.
- Developed business process automations using Microsoft Graph API and SQL to streamline system integrations.

Undergraduate Research Assistant Co-op @ University of Guelph - Guelph ON *Apr. 2026 - Sep. 2026*
- Developed and evaluated NLP pipelines for document extraction, table reconstruction, and OCR validation
- Used Python to parse benchmark dataset statistics, disclosure distributions, and concept-level summaries
- Architected rigorous, domain-specific ESG evaluation frameworks to assess evidence-grounded LLM reasoning.

Campus Organizer @ Google Developer Groups on Campus - Guelph ON *Sep. 2025 - May. 2026*
- Selected by Google as the University's Campus Lead, managing a developer group with 647 students
- Ideated and pitched initiatives to Google, securing funding for 5 technical workshops and a 24-hour hackathon
- Oversaw the hiring and coordination of 13 personnel in Financial, Technical, Marketing, and Events departments

Data Entry Assistant @ Hawkey Church Management - Mount Forest ON *May. 2025 - Sep. 2025*
- Utilized Excel VBA and Google Maps API to generate graphical visualizations of the companies construction sites
- Wrote Python scripts to modernize databases, adding Optical Character Recognition on legacy documents
- Designed organizational tools for tracking and managing construction projects, improving efficiency by 70%

Projects

RAG LLM Drone Chatbot · Python, Flask, ChromaDB, vector databases, REST API, Google Gemini API *Feb. 2026*
- Built a Flask-based web service and REST API that enforced source-cited answers and rate limits for safety.
- Created a RAG pipeline with chunking and a vectorized DB to serve source-cited answers on Canadian drone rules.
- Implemented an AutoImmunePipeline to detect data-poisoning and jurisdiction mismatches.

Hybrid Quantum-Classical ML Classifer · Python, Pennylane, Quantum Computing, MLflow *Feb. 2026*
- Developed a hybrid Quantum-Classical ml classifier, achieving a 7% improvement in accuracy over classical models
- Utilized pennylane to design and implement quantum circuits that processed data in a non-euclidean space
- Applied standard research software development policies including reproducibility, documentation, and git control

Safe Biking Routes App · React, TypeScript, Python, Axios, Flask *Jun. 2025*
- Developed a customized A* algorithm to create personalized route pathing, creating an optimal path that balanced the safety and expected time of arrival, optimizing the quality of experience for a diverse userbase.
- Implemented data-driven route pathing, avoiding historically dangerous intersections and prioritizing cyclist safety

Education

University of Guelph Guelph, ON
B.Comp. in Computer Science Co-op, Stats minor *Sep. 2024 - Apr. 2028*
Relevant Coursework: Linear Algebra, Calc III, Statistics, Data Structures and Algorithms

Extracurriculars and Volunteering

Hackathon Coordinator @ Google Developer Groups on Campus *Sep. 2024 - May 2025*
- Ran an event with 260 accepted applicants, coordinated event logistics, and secured $5000 in funding
- Co-directed several technical events including Webdev, React.js, Cloud Computing and Firebase Workshops
- Comprehensively trained and led a team of 17 volunteers, ensuring coordination and success of events

Computer Science Club Leader *Sep. 2023 - Jun. 2024*
- Mentored advanced students by teaching concepts such as graph traversal, dynamic programming and recursion
- Fostered a conducive learning environment by promoting diversity and equality amongst members and attendees
- Introduced 40+ members to Python fundamentals, mentoring them in creating their own passion projects